Solve the follwing system:

$$\begin{cases} \log_{10} x + \log_{10} y = 2 \\ 2x - y = -10 \end{cases}$$

$$\log_{10} x + \log_{10} y = 2 \Rightarrow \frac{\ln x}{\ln 10} + \frac{\ln y}{\ln 10} = 2$$

$$\Rightarrow \frac{\ln(xy)}{\ln 10} = 2 \Rightarrow \ln(xy) = 2\ln 10$$

$$e^{\ln(xy)} = e^{\ln(10)^2} \Rightarrow xy = 100$$

$$\begin{cases} xy = 100 \\ 2x - y = -10 \end{cases} \Rightarrow -y = -2x - 10$$

$$y = 2x + 10$$

$$x(2x+10) = 100$$

$$2x^2 + 10x = 100$$

$$2x^2 + 10x - 100 = 0$$

$$\boxed{x = 5} \quad , \quad x = -10 \Rightarrow \text{rej}$$

$$\log_{10}(x) > 0$$

$$\Rightarrow 5y = 100$$

$$y = \frac{100}{5} = 20$$

$$\boxed{y = 20}$$

$$S : \{(5, 20)\}$$

Day 6